

SOS 2026 - Markov Chains

Plan of Action

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Timeline:

Week1: Laying the mathematical groundwork, Review of probability axioms, conditional probability, total probability, and Bayes' Theorem, Understanding Bernoulli, Binomial, Poisson, Uniform, and Normal (Gaussian) distributions. Concepts of Expectation, Variance, and Covariance.

Week2: Introducing state transitions and memorylessness. Introduction to generic stochastic processes and the definition of the "Markov Property." Visualizing simple Markov Chains using state transition diagrams. Building the fundamentals of Markov Chains. The Chapman-Kolmogorov equations and multi-step transition probabilities.

Week3: Learning about DTMCs. Analyzing the structural and long-term properties of DTMCs. State classification: transience, recurrence, and communicating classes. Stationary (invariant) distributions, limiting probabilities, and the Ergodic Theorem.

Week4: Introduction to Continuous-Time Markov Chains (CTMCs) and the Q-matrix. Understanding the Poisson Process as in CTMC: arrival times, interarrival times, and spatial randomness. Compile all mathematical notes, probability proofs, and computational simulations into the formal midterm report.

MidTerm Submission

Week5: Introduction to MDPs. Defining the core components of MDPs: States, Actions, Transition Probabilities, and Rewards. Understanding Policies (deterministic vs. stochastic) and cumulative Return.

Week6: Understanding Reinforcement learning fundamentals. Algorithmic solutions for decision-making. Solving known MDPs via Dynamic Programming. Introduction to Model-Free Reinforcement Learning.

Week7: Hands-on implementation of RL algorithms. Build, train, and optimize a Q-Learning agent from scratch. Plot its learning curve as it converges on an optimal strategy within a simulated environment.

Week8: Buffer week for revision, revisiting complex proofs, remaining work or refinements in the RL Agent and final documentation.

EndTerm Submission

References:

- Book Markov Chains by J. R. Norris
- Book Introduction to Probability Models by Sheldon Ross
- Book Markov Chains and Mixing Times by Levin, Peres & Wilmer
- Book Reinforcement Learning: An Introduction by Sutton & Barto
- 3Blue1Brown's Probability playlist on YouTube